

PRINCIPLES OF DISCRETE APPLIED MATHEMATICS

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LECTURE 1: THE PIGEONHOLE PRINCIPLE

THE PIGEONHOLE PRINCIPLE says that if there are more objects than boxes, some box must hold at least two objects. Its main use is to prove that a repetition must occur, even when we do not know where.

For example, a drawer contains socks of two colors. Picking three socks in the dark guarantees a matching pair: three socks cannot all have different colors when only two colors are available.

1.1 Statement

Definition: If m objects, called PIGEONS, are placed into n boxes, called PIGEONHOLES, and $m > n$, then at least one box contains more than one object. In other words, some two objects share a box.

A stronger form tells us how many objects must share a box.

Theorem 1.1 [PIGEONHOLE PRINCIPLE, STRONG FORM]. *If m objects are placed into n boxes, then at least one box contains at least*

$$\left\lceil \frac{m}{n} \right\rceil$$

objects.

Here $\lceil \cdot \rceil$ denotes rounding up to the next integer, while $\lfloor \cdot \rfloor$, used later, denotes rounding down. The basic form is the case $m > n$, where $\lceil m/n \rceil \geq 2$.

Note. The hard part is choosing the objects and boxes. Once those choices are clear, count them and compare the totals.

1.2 Two Vertices of Equal Degree

A GRAPH G consists of a set V , whose elements are called VERTICES, and a set E of pairs of vertices, whose elements are called EDGES. The DEGREE of a vertex v is the number of edges containing v .

Theorem 1.2. *In any finite simple undirected graph with at least two vertices, there are two vertices of equal degree.*

At a party with n people, this says that two people know the same number of people at the party.

Proof. Let the graph have n vertices. Every degree is an integer between 0 and $n - 1$, which gives n possible values for n vertices, so counting alone does not force a repeat. Suppose for

contradiction that all degrees are distinct. Then every possible value occurs exactly once. In particular, some vertex v has degree 0, while some vertex w has degree $n - 1$ and is adjacent to every other vertex.

These two vertices cannot coexist. If the edge (v, w) is present, then v has a neighbor and does not have degree 0. If the edge is absent, then w is not adjacent to every other vertex and does not have degree $n - 1$. Either case is a contradiction, so two degrees must coincide. $\square$

Note. The pigeons are the n vertices and the pigeonholes are the n possible degrees, so the counts initially match. The proof rules out simultaneous use of degrees 0 and $n - 1$, leaving only $n - 1$ usable holes for n pigeons.

1.3 Equal Sum Subsets

Here the principle does something less obvious.

Theorem 1.3. *Among any 30 seven-digit numbers, there exist two disjoint nonempty subsets with the same sum.*

Proof. Take the nonempty subsets as pigeons. There are $2^{30} - 1$ of them. Take the possible subset sums as pigeonholes. Each number is less than 10^7 , so the sum of at most thirty numbers is at most $3 \cdot 10^8$, which is smaller than $2^{30} - 1 \approx 10^9$. There are therefore more subsets than achievable sums, so two distinct subsets A and B satisfy

$$\sum_{x \in A} x = \sum_{x \in B} x.$$

The subsets need not initially be disjoint, but their common elements can be discarded. Put $A' = A \setminus B$ and $B' = B \setminus A$. Removing the same quantity $\sum_{x \in A \cap B} x$ from both sides preserves equality, while A' and B' are disjoint. Since A and B were distinct, at least one of A' and B' is nonempty; equality of sums forces the other to be nonempty as well. $\square$

Note. The proof tells us that the subsets exist, but not how to find them quickly. With many more numbers, finding the matching subsets can still be expensive.

1.4 Monotone Subsequences

A **SUBSEQUENCE** of $a_1, a_2, \dots, a_k$ is obtained by deleting some terms while preserving the order of the rest. It is **MONOTONE** when it is entirely increasing or entirely decreasing. Long sequences of distinct real numbers must contain long monotone subsequences.

Consider the length-12 sequence

$$4, 3, 2, 1, 8, 7, 6, 5, 12, 11, 10, 9,$$

plotted in figure 1. The terms 4, 8, 11 form an increasing subsequence of length 3, while 4, 3, 2, 1 form a decreasing subsequence of length 4. Neither can be improved within this sequence. With $m = 3$ and $n = 4$, it has exactly $mn = 12$ terms, one short of the $mn + 1 = 13$ terms that force a longer subsequence.

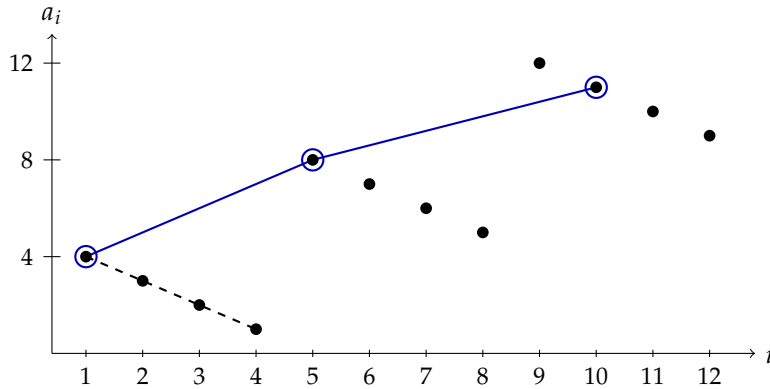


Figure 1: The sequence 4, 3, 2, 1, 8, 7, 6, 5, 12, 11, 10, 9 plotted as points (i, a_i) . The solid path marks the increasing subsequence 4, 8, 11 and the dashed path the decreasing subsequence 4, 3, 2, 1; neither can be lengthened.

Theorem 1.4. Any sequence of $mn + 1$ distinct real numbers $a_1, a_2, \dots, a_{mn+1}$ has an increasing subsequence of length $m + 1$ or a decreasing subsequence of length $n + 1$.

Proof. Let $t(i)$ denote the length of the longest increasing subsequence ending at a_i . If $t(i) > m$ for some i , an increasing subsequence of length at least $m + 1$ exists and we are done. Assume instead that $t(i) \in \{1, 2, \dots, m\}$ for every i .

Apply the pigeonhole principle to the $mn + 1$ indices and the m possible values of t . Some value s satisfies $t(i_j) = s$ for at least $n + 1$ indices

$$i_1 < i_2 < \dots < i_{n+1}.$$

We claim that

$$a_{i_1} > a_{i_2} > \dots > a_{i_{n+1}},$$

which is a decreasing subsequence of length $n + 1$. Otherwise some consecutive pair satisfies $a_{i_j} < a_{i_{j+1}}$. The increasing subsequence of length s ending at a_{i_j} could then be extended by $a_{i_{j+1}}$, producing an increasing subsequence of length $s + 1$ ending there. This contradicts $t(i_{j+1}) = s$. $\square$

Exercise. For any m and n , construct a sequence of mn distinct numbers with no increasing subsequence of length $m + 1$ and no decreasing subsequence of length $n + 1$. This shows the theorem

is best possible.

1.5 Twenty Questions

One player thinks of an object and a second player asks yes-or-no questions until they can name it. How many distinct objects can be distinguished with twenty questions?

The strategy may be adaptive: the k th question can depend on the preceding $k - 1$ answers. Restrict the guesser to a deterministic strategy, so that the same answer history always produces the same next question. Such a strategy is a binary tree (figure 2): each internal node is a question, each branch an answer, and each leaf a complete answer history.

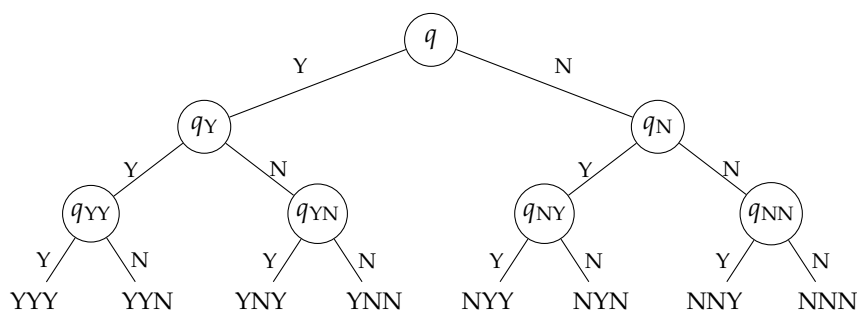


Figure 2: A deterministic adaptive strategy with three questions. The question asked next depends only on the answers so far, so each object that reaches a leaf is identified by its answer history. With twenty questions the tree has 2^{20} leaves.

Even when questions depend on earlier answers, the guesser must distinguish every pair of objects. If two objects produced the same answers throughout, the strategy would ask the same questions for both and finish in the same state, forcing an incorrect guess for at least one of them.

Thus each object determines a unique sequence of twenty yes-or-no answers. Take objects as pigeons and answer sequences as pigeonholes. There are 2^{20} sequences, so at most 2^{20} objects, a little over one million, can be distinguished. This bound is attained by a strategy that separates 2^{20} objects in twenty questions.

Note. If every object must be identified without error, random choices cannot increase this counting bound. Allowing a chance of error would change the problem.

1.6 Twenty Questions with One Lie

Now allow the answerer to lie once, a variation apparently first considered by Stanislaw Ulam. How many objects can the guesser distinguish?

For a fixed object, the answerer may tell the truth throughout or lie on any one of the twenty questions. Once the lie is used, all remaining answers are forced. Hence each object is compatible

with 21 distinct answer sequences. With t objects there are $21t$ pigeons and still 2^{20} pigeonholes.

If $21t > 2^{20}$, two pigeons share a hole. They cannot represent the same object because its 21 compatible sequences are distinct. Therefore some answer sequence is compatible with two objects, so the guesser cannot distinguish them. The maximum is bounded by

$$t \leq \left\lfloor \frac{2^{20}}{21} \right\rfloor = 49932.$$

Note. This is an upper bound; whether a strategy attains it is a separate question. For some numbers of questions such a strategy exists, and otherwise one can come reasonably close. The relevant tools appear later with error-correcting codes: a lie is a transmission error, and the questions must be robust to one such error.

Source: MIT 18.200 Principles of Discrete Applied Mathematics, Spring 2024, [Lecture 1 notes](#) by Ankur Moitra.